

Research Article

**Gamification in higher education - A study on
Instructors' and Students'
Perception in Gulf Region.**

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Abstract:

Increased utilization of Gamification has been facilitated by the ground-breaking technological advancements in the recent years. This empirical study aimed to determine the instructors' and students' perception on gamification in the Gulf Region. The current study also explored the differences in perception on gamification based on age, gender, nationality and level of education. Mixed methods were chosen in this study, whereby researchers collected and analyzed both quantitative and qualitative data. Stratified sample method was used to collect data from 800 students and 160 faculty from eight universities in UAE, Oman, Qatar and Bahrain. Descriptive statistical analysis was used to find the mean and standard deviation of each of the variable. Chi-square test and ANOVA were used to find out the mean difference between the groups. The research findings will be helpful for educators and policy makers in designing more effective teaching pedagogies and also help in the identification of training needs for faculty development to ensure more effective utilization of gaming strategy, as an effective strategy to enhance students' attention in the classroom.

Keywords: Gamification, perception, technological advancement, teaching pedagogies, attention, student engagement, AI

Gamification, Higher Education, Student Engagement, AI, Technology in Education

Introduction

Gamification is a very recent development that refers to the use of gaming approach to create a more effective and engaging learning environment. Gamification is rapidly gaining widespread acceptance among students and teachers alike, mainly as it stimulates the mind to think and increases positive classroom interaction thereby accomplishing learning in a less stressful and fun way. Yu, Gao and Wang (2020), defined gamification as a "series of design principles, processes and systems used to influence, engage and motivate individuals, groups and communities to drive behaviors and effect desired outcomes."

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The methodology of teaching using games is one of the most effective pedagogical methods, especially within the stages of feedback and testing students' initial understanding of the given content.

With Gamification, learning becomes highly effective as students are increasingly engaged in the classroom. Gamification provides a more effective supplement to the existing active learning strategies. It also serves as an effective and innovative pedagogical tool that harnesses the immense potential of recent technological development in achieving curriculum related goals.

Globally, this decade witnessed the emergence of 'Gamification' to enhance the level of students' engagement in the classroom. Academics have been continuously analyzing the factors affecting students' learning and accomplishment of learning outcomes. Swift transition to online or hybrid was necessitated by social distancing norms due to the Covid19 global pandemic. Educational institutions across the globe showed remarkable resilience in this transition from face to face to hybrid or online teaching with gamification strategy emerging as an effective classroom engagement method in the process.

The aim of this study is to analyze the perception of students and instructors on gamification in the teaching and learning process. Quantitative and qualitative research approaches will be applied to understand and analyze the perception of student and instructors about gamification. The research findings will be employed to mentor teachers and policy makers in Abu Dhabi in particular and UAE and the region, in general, to adapt and implement gamification as a pedagogical tool.

The potential of employing wide range of technology in education settings is immense and it could have been harnessed to a more significant level than the prevailing level. The limitation was perhaps imposed due to the inhibiting impact of absence of standardized and universally acceptable policy framework on the acceptance of technology in general and gamification in particular. Globally, the social distancing norms that have been put in place since March 2020 due to Covid19 have served as significant catalyst in bringing about the much desired paradigm shift in the utilization of gamification. Studies, like the current one, will certainly go a long way in creating the much needed and desired awareness of gamification that will enhance student engagement with the ever increasing use of technology in learning.

Methodology

Mixed method will be chosen for this study, whereby researchers will collect and analyze both quantitative and qualitative data within the same study. Researchers will also obtain approval from institutional review board before data collection.

Sample:

The sample of this study will comprise 160 Faculties and 800 students (Male & Female) selected randomly from five Universities in UAE, Oman, Qatar and Bahrain.

Gamification

Teachers play an important role in introducing pedagogical innovations in the classroom, particularly technological innovations (Ketelhut & Schifter, 2011), so teachers will play an important role in adopting the use of gamification in their courses based on their attitude toward gamification. To gain a better understanding of teachers' actual use of gamification, we must first examine teachers' attitudes toward gamification, as attitude is a strong predictor of an individual's behaviour. The actual use of gamification by teachers is analysed in relation to gender, age, discipline which they teach and the level in which they teach i.e., School, or higher education level.

Attitude towards Gamification

The term "attitude" refers to an individual's level of evaluative affect toward a specific behavior (Fishbein & Ajzen, 1975). As a result, attitude is a significant predictor of an individual's intention to perform the target behaviour. Attitude toward gamification can be defined as the evaluative affect toward using gamification as a teaching tool. In order to gain a better understanding of teachers' attitudes toward gamification, we pose the following research questions: RQ-1, I prefer to use online games in class in comparison to the use of only traditional methods, RQ-2, The online games can be easily adapted to every subject, RQ-3, The online games help students to understand the basic and advanced concepts clearly, RQ-4, I prefer online games because it enables teachers to guide and reward students.

Use of gamification

Actual gamification use can provide useful insights into the role of gamification as a teaching methodology in classrooms. We pose the following research question in order to gain a better understanding of teachers' actual use of gamification in classroom transactions RQ-5, I think games facilitate student's engagement in the active learning process in classroom RQ-6, I prefer to use online games as it makes learning more interesting RQ-7, Online games enable students to maximize their potential.

Gender

In contexts of gamification as a teaching tool, gender has been found to play an important moderating role. Despite the fact that gender differences appear to be diminishing among younger generations (ESA, 2015), boys typically have more positive attitudes toward video gaming than girls (Bonanno & Kommers, 2008). Gender differences have also been observed during educational games played in class (Bressler & Bodzin, 2013). On the basis of these researches, the investigator taken effort to study the gender differences in the present study.

Age

Age has been discovered to be an important variable in education, as older teachers are thought to have more experience than younger ones. Nonetheless, previous research suggests that age can have a negative impact on older teachers' use of Information and Communication Technologies (ICT). According to Goodwyn, Adams & Clarke (1997), ICT is generally viewed as a threat and a source of great anxiety among older teachers. Because gamification is a pedagogical innovation that is typically associated with technology (e.g., video games), we believe that teachers' attitudes and actual use of gamification may be influenced by their age. As a result, the researcher analysed the data on the basis of participants' age.

Level of teaching of the Teacher

The researcher wanted to the study whether there is any significant relationship between the attitude of teachers towards gamification vary with the level in which they teach. Hence the analysis is done in terms of the level of teaching of participants.

Results of the Faculty perception

Demographic Details of the sample

1. Teaching Faculty

The study includes 302 Teaching faculty. The demographic details include the gender, age, level of teaching and subjects of teaching. The male participants include 25.50% and female participants include 74.50%. The respondents were again categorized into 5 age groups 25 to 30, 31 to 35, 36 to 44, 45 to 50 and 50 and above the maximum number of participants belongs to which are 25-30.

The participants are analysed with respect to the level of teaching and the subject of teaching The maximum number of participants are from the school as well as from the miscellaneous categories which includes online teaching, tutoring and the freelance working. The subject of teaching of the participants were also considered under the demographic details of the teaching faculty. The data shows that 36% of the participants were from the social sciences, 28% belong to mixed category which includes the teachers who teaches interdisciplinary programs.

2. Student Participants

The study includes 420 students. The demographic details include the gender, age, year of study, nationality, and Discipline. The male participants include 30.47 % and female participants include 69.52%. The maximum respondents were from the 18-20 years followed by 21-22 and 25 years. The students from the 3rd and 4th year of their studies. into 5 age groups 25 to 30, 31 to 35, 36 to 44, 45 to 50 and 50 and above the maximum number of participants belongs to which are 25-30. The student community from UAE represent a cosmopolitan distribution wherein students from different countries come for their higher education. The student participants were also analysed with respect to their

academic discipline. The data shows that 32.85% of the participants were from the Engineering, 21% belong to sciences and 17% from the mixed category which includes the students who studies interdisciplinary programme.

Faculty perception on student's engagement and Motivation construct of gamification

There are 3 research questions which analyses the component of gamification student engagement and motivation. The table 1 shows the response in terms of gender variations on student engagement and motivation. The chi-square statistic is 0.2453. The p-value is .884567. The result is not significant at $p < .05$. Hence there are no significant difference in relation to gender on the component of gamification - student engagement and motivation.

I think games facilitate student's engagement in the active learning process in the classroom

I prefer to use online games as it makes learning more interesting

Online games enable students to maximize their potential

Table 1. Gender and gamification:

	Category 1	Category 2	Category 3	Row Totals
Group 1	178 (175.42) [0.04]	23 (24.73) [0.12]	30 (30.85) [0.02]	231
Group 2	510 (512.58) [0.01]	74 (72.27) [0.04]	91 (90.15) [0.01]	675
Column Totals	688	97	121	906 (Grand Total)

The chi-square statistic is 0.2453. The p-value is .884567. The result is not significant at $p < .05$.

The analysis of each of the research questions from the category shows that the teachers who have a positive attitude towards gamification prefer it because they consider the gamification increases the student engagement in the active learning process $n=286$ and mean scores-768. The factor 'learning makes more interesting' $n=239$ and mean scores 717. There are good number of teachers (Mean= 579 and $n=193$) recommend gamification as they found it helpful in maximizing the learning potential of students.

Table 2. Faculty perception of gamification:

	Category 1	Category 2	Category 3	Row Totals
Group 1	296 (283.13) [0.59]	57 (54.65) [0.10]	49 (64.22) [3.61]	402
Group 2	621 (633.87) [0.26]	120 (122.35) [0.05]	159 (143.78) [1.61]	900
Column Totals	917	177	208	1302 (Grand Total)

The chi-square statistic is 6.2117. The p-value is .044787. The result is significant at $p < .05$.

The analysis of the research questions on gamification in the classroom context individually showing the following results. The teachers found gamification most useful tool for helping their students understand the basic and advanced concepts in the subjects $n=225$ with mean scores of 675. Teachers also responded that the gamification allows them to guide and reward students easily ($n=211$ Mean scores 633). Overall, the analysis shows that teachers hold a positive attitude towards gamification due to its adaptability to different subjects and higher student engagement.

The results show that majority of the teachers irrespective of the demographic variations have a favourable attitude towards gamification $n=194$, Mean =582, whereas $n=62$ participants do not have a positive attitude towards gamification and there 15% of the respondents who are not sure about the potentials of online gamification as potential instructional tool.

Student Perception on the construct of motivation/involvement in learning

There are 3 research questions which analyses the construct of gamification student involvement and motivation. The table 3 shows the response in terms of gender variations on student engagement and motivation. The chi-square statistic is 3.0213. The p-value is .220769 The result is not significant at $p < .05$. Hence there are no significant difference in relation to gender on the component of gamification - student involvement and motivation.

Table 3. Student engagement:

	Agree	Disagree	Not sure	Row Totals
Female	228 (222.48) [0.14]	27 (31.98) [0.78]	37 (37.54) [0.01]	292
Male	92 (97.52) [0.31]	19 (14.02) [1.77]	17 (16.46) [0.02]	128
Column Totals	320	46	54	420 (Grand Total)

The chi-square statistic is 3.0213. The p-value is .220769. The result is not significant at $p < .05$. The chi-square for the preference in the usage of gamification as individual or group were 3.4789 $p=.17562$ $p<.05$ and is 2.704. The p-value is .258718. The result is not significant at $p < .05$. consecutively. Thus it is clear from the analysis that there are no significant gender variations on the construct of student motivation/involvement in learning.

Student Perception on the construct of game interface

There are 6 research questions which analyses the construct of game interface. The table 4 shows the response in terms of gender variations on game interface which includes colour variations, navigations, designs and layouts. The chi-square statistic is 8.8203. The p-value is .012153The result is significant at $p < .05$. Hence there are significant difference in relation to gender on the construct of game interface.

Table 4. construct of game interface:

	Agree	Disagree	Not sure	Row Totals
Female	1386 (1387.70) [0.00]	130 (144.61) [1.48]	236 (219.70) [1.21]	1752
Male	610 (608.30) [0.00]	78 (63.39) [3.37]	80 (96.30) [2.76]	768
Column Totals	1996	208	316	2520 (Grand Total)

The chi-square statistic is 8.8203. The p-value is .012153. The result is significant at $p < .05$.

Student Perception on the construct of usability of online games for their easy access

There are 4 research questions which analyses the construct of usability of online games for their easy access. The table 5 shows the response in terms of gender variations on usability of games. The chi-square statistic is 31.8192. The p-value is < 0.00001 . The result is significant at $p < .05$. Hence there are significant difference in relation to gender on the construct of game interface.

Table 5. Construct of usability of online for their easy access

	Agree	Disagree	Not sure	Row Totals
Female	845 (831.19) [0.23]	108 (140.97) [7.71]	239 (219.84) [1.67]	1192
Male	346 (359.81) [0.53]	94 (61.03) [17.82]	76 (95.16) [3.86]	516
Column Totals	1191	202	315	1708 (Grand Total)

The chi-square statistic is 31.8192. The p-value is < 0.00001 . The result is significant at $p < .05$.

Student Perception on the construct of cognitive development of online games

There are 3 research questions which analyses the construct of cognitive development of online games. The table 6 shows the response in terms of gender variations on cognitive development of online games. The chi-square statistic is 29.8031. The p-value is < 0.00001 . The result is significant at $p < .05$. Hence there are significant difference in relation to gender on the construct of game interface.

Table 6. The construct of cognitive development of online games

	Agree	Disagree	Not sure	Row Totals
Female	714 (688.98) [0.91]	48 (72.30) [8.17]	114 (114.71) [0.00]	876
Male	277 (302.02) [2.07]	56 (31.70) [18.64]	51 (50.29) [0.01]	384
Column Totals	991	104	165	1260 (Grand Total)

The chi-square statistic is 29.8031. The p-value is < 0.00001 . The result is significant at $p < .05$.

Conclusion

This study aimed at exploring features for perspective towards using gamification techniques and online learning. The objectives of the study included on an exploratory survey that had sections including the effects of incorporating game elements in the classroom context and games for student engagement and motivation. The results of the present study have revealed that most of the participants have a positive attitude towards the incorporation of games in the learning environment and this supports the earlier research. This study also revealed that teachers had a positive attitude because they realized that games can be adaptable in classroom situations and maximizes students' potential and makes learning more interesting. The gamification increases the student engagement in the active learning process.

Among various factors the teacher participants believe that gamification is adapted to every subjects and it helps students to understand the basic and the advanced concepts and it always help the teachers to guide and reward students in comparison to the use of our traditional methods. The results also show teachers find gamification acts as a good reinforcement as it provides immediate feedback in terms of right or wrong or reward points, stars and trophies etc. This motivates students and improve student active engagement. The gamification sustains the motivation so high that it engages the students in completing complex activities and discussions without any hindrances.

In conclusion, Gamification represents an innovative and engaging instructional tool to motivate students and enhance their learning process. Through this exploratory study the researcher throw light on teachers' attitude towards gamification and the outcome of the study show that teachers hold a positive attitude towards gamification.

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